

The Space Cost of Shor's Algorithm

How much does the Quantum Phase Estimation register weigh



Shor's Algorithm: the Full Pipeline

1

Order-finding reduction

Factoring N reduces to finding the order r of a random a coprime to N : $\gcd(a^{r/2} \pm 1, N)$ yields a non-trivial factor with probability $\geq 1 - \frac{1}{2^{k-1}}$.

2

Quantum Phase Estimation

A QPE circuit estimates the eigenphase $\theta = k/r$ of the operator U_a on a counting register of t qubits.

3

Continued fractions

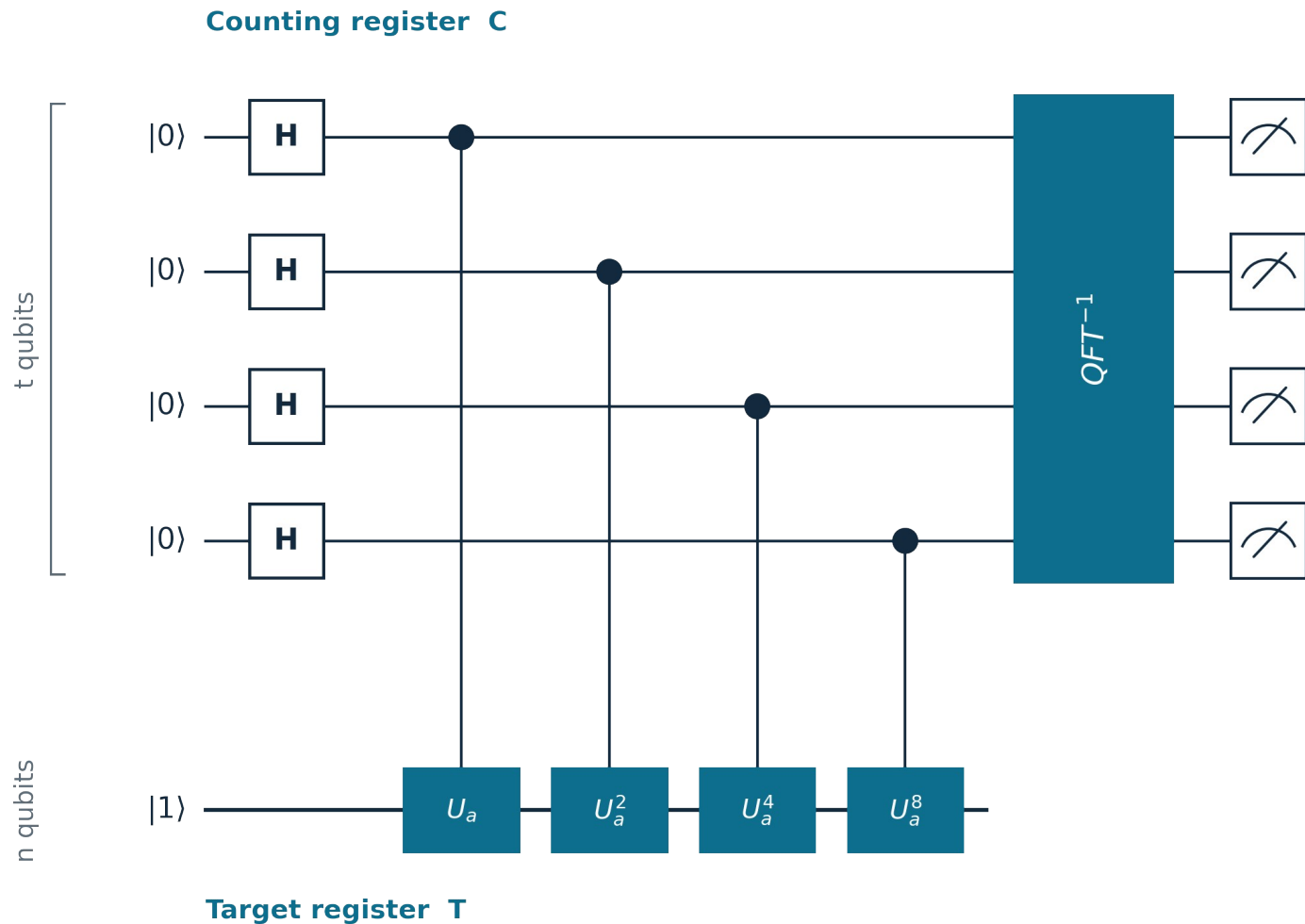
The measured phase $y/2^t$ is expanded as a continued fraction to recover the best approximation k/r , with denominator $< N$.

4

Euclidean algorithm

Once r is known, $\gcd(a^{r/2} - 1, N)$ is computed classically in polynomial time to extract the factors of N .

Inside the QPE Circuit



The operator U_a

$U_a |x\rangle = |ax \bmod N\rangle$
 is a permutation, hence unitary.
 Its eigenvectors are
 $|u_j\rangle = (1/\sqrt{r}) \sum_k w_r^{-jk} |a^k \bmod N\rangle$,
 $j = 0, \dots, r-1$,
 with eigenvalue $w_r^j = e^{2\pi i j/r}$.

Why start from $|1\rangle$

$|1\rangle = (1/\sqrt{r}) \sum_j |u_j\rangle$
 is an equal superposition of all r
 eigenvectors of U_a —
 we cannot prepare a single $|u_j\rangle$
 without already knowing r ,
 but $|1\rangle$ needs no such knowledge.

Order finding uses two registers

- **Target register T** — $n = \lceil \log_2 N \rceil$ qubits, holds the state $|a^k \bmod N\rangle$
- **Counting register C** — t qubits, reads the phase $\theta = k/r$ and nothing else

How large must C be?

Register C • t qubits



QFT†

Register T • n qubits



total width = $t + n$

From phase to order: continued fractions

Measuring C returns $y/2^t \approx k/r$ with k uniform in $\{0, \dots, r-1\}$.

Continued fractions reconstruct k/r exactly — but only if the estimate is precise enough:

$$\left| \frac{y}{2^t} - \frac{k}{r} \right| < \frac{1}{2r^2}$$

- The phase must be resolved with error $< 1/(2r^2)$.
- Register C sets the resolution: grid spacing 2^{-t} .
- With too few bits, multiple phases k/r land on the same measured value and continued fractions get it wrong.

Example ($N = 21$, $t = 6$)

measured $y/2^t = 11/64 \approx 0.172$

convergents: 0, $1/5$, **$1/6$** $\rightarrow r = 6$

the first denominator $< N$ is chosen.

The space cost: $t \approx 2n$

- Since $r < N$, it suffices to require
 $|y \cdot 2^t - k \cdot r| < 1/(2N^2)$
- which requires $t \gtrsim 2 \log_2 N + 1 \approx 2n$

The register that only reads the phase is twice as large as the one holding the number.

- Total width $\approx 3n$ versus $\log_2 N$ bits of useful information.

register	size	role
target T	n	holds the number
counting C	$\approx 2n$	phase precision
TOTAL	$\approx 3n$	



two-thirds of the width is just phase precision

2×

C versus T

3n

total width

The experiment

Only t is varied, and from the exact distribution of register C two probabilities are measured:

$P_{\text{resolve}}(t)$

phase measured within $1/(2r^2)$ of a true k/r .

purely spatial metric: «did the register have enough bits?»

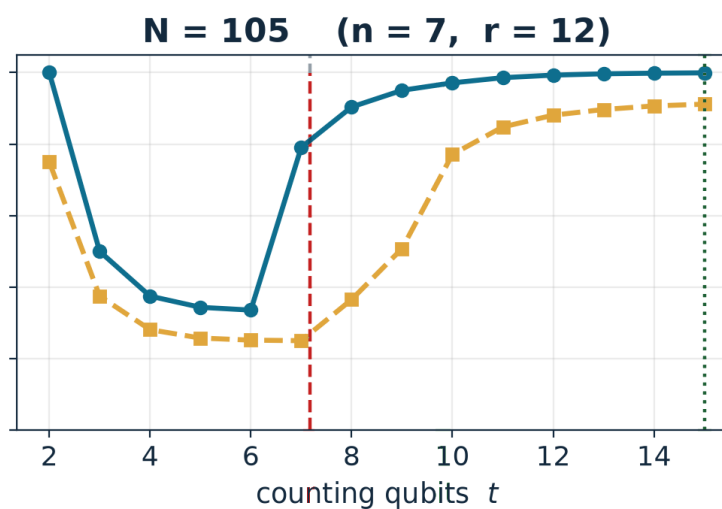
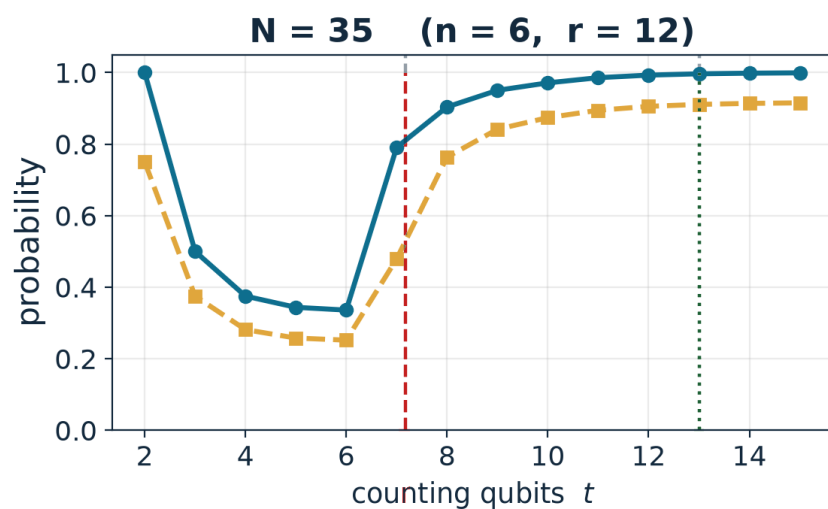
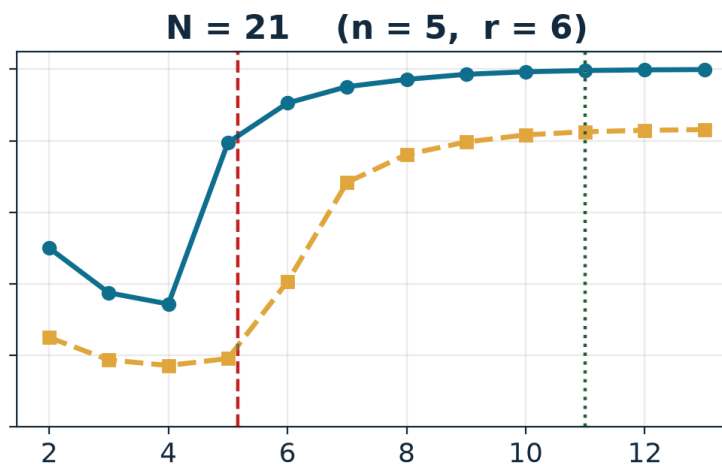
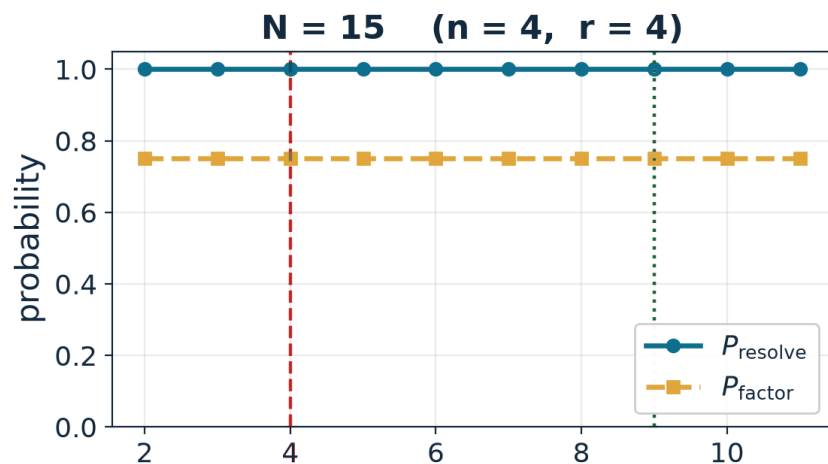
$P_{\text{factor}}(t)$

a single shot yields a non-trivial factor of N .

includes the numerical overhead: $k = 0$, r odd, $\gcd(k, r) > 1$.

- The gap $P_{\text{resolve}} - P_{\text{factor}}$ is **fixed** (theorem: $\text{success} \geq 1 - \frac{1}{2}^{k-1}$, k = number of distinct primes);
- while the point where P_{resolve} rises is the **space requirement**.
- Target: $N = 15, 21, 35, 105$ with base $a = 2$. Simulator: PennyLane default.qubit, analytic mode.

Success vs. register size



The «knee»

P_{resolve} rises around $t \approx 2 \log_2 r$, well inside the worst-case bound $2n+1$.

Plateau

P_{factor} plateaus below 1: this is the numerical overhead, independent of t .

N = 15

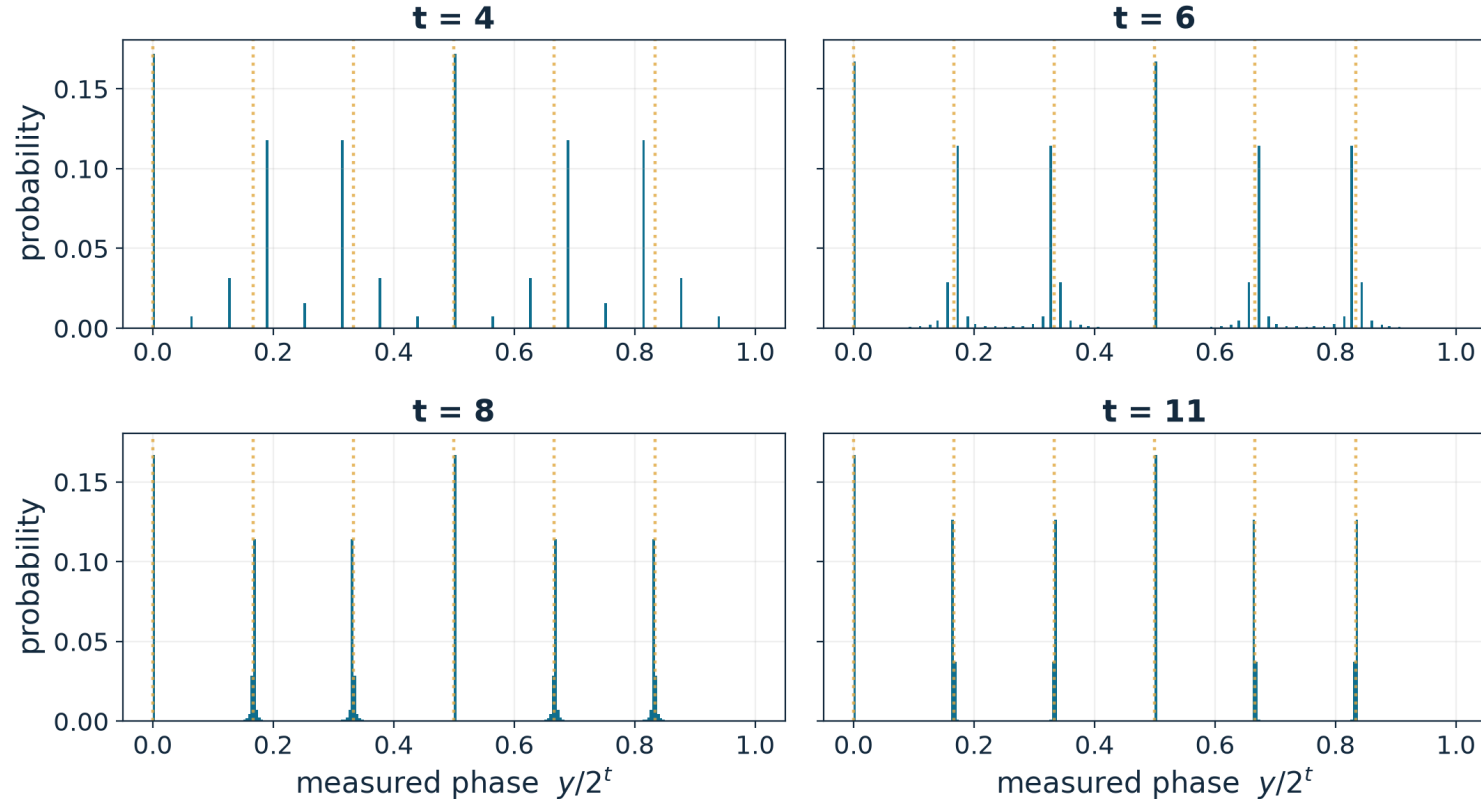
flat curves: $r = 4$ is a power of 2.

red dashed $t = 2 \log_2 r$ (instance)

dark green dotted $t = 2n + 1$ (worst-case bound)

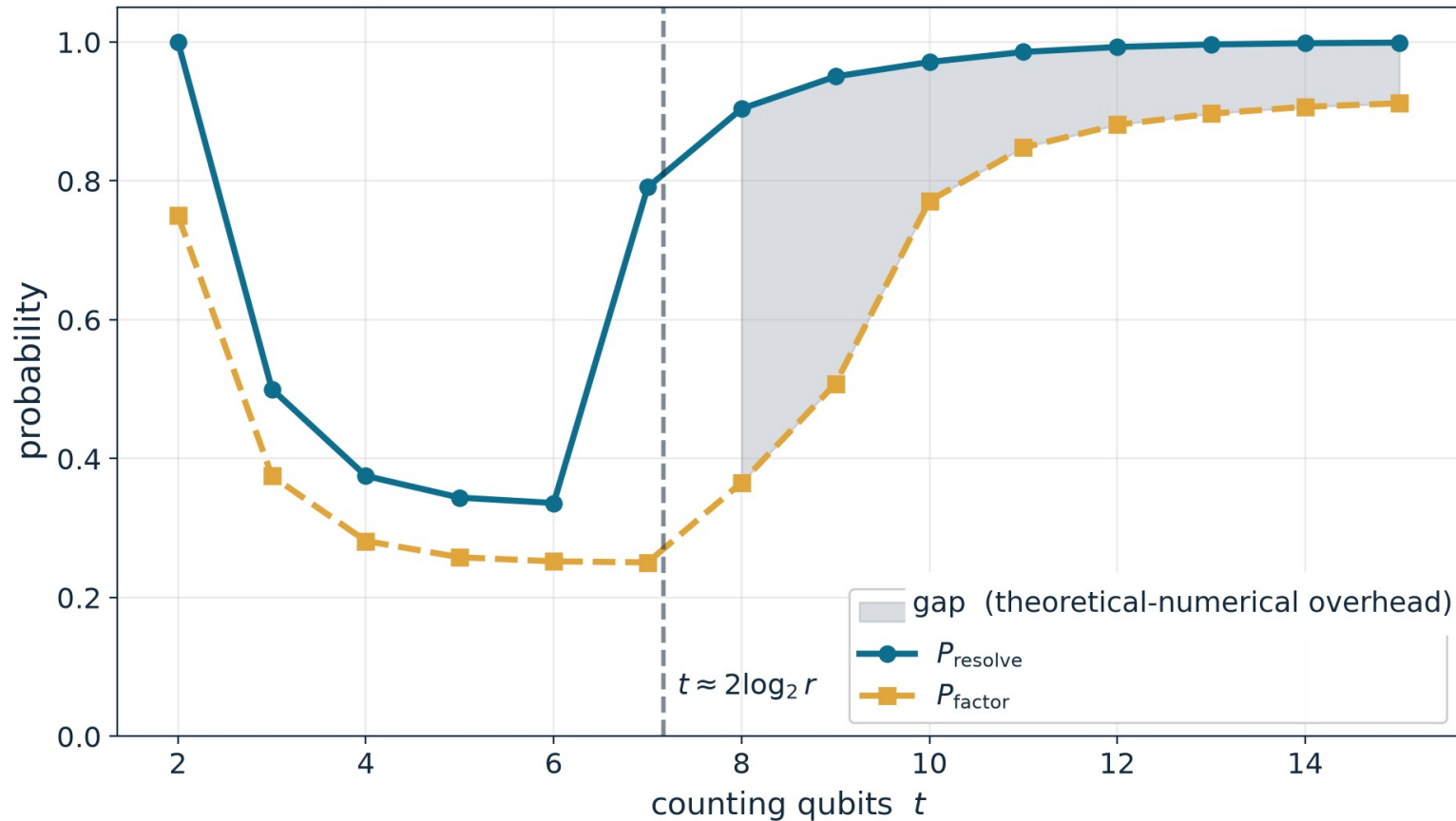
$N = 15$ is deceptive — the non-monotonic dip

$N = 21$: peaks align with $k/6$ (amber lines) only as t grows



- $N = 15$ has $r = 4 = 2^2$: the phases $k/4$ are exact in 2 bits $\rightarrow$ success already at $t = 2$.
- With t tiny, the only representable values $y/2^t$ coincide by aliasing with true k/r values.
- As soon as t is enough to sample phases that are NOT k/r but not enough to resolve them, success collapses (minimum at $t \approx n$).

Space requirement vs. numerical overhead



Rise of P_{resolve}

= the register has reached $\approx 2 \log_2 r$ bits. This is the SPACE cost.

Residual gap

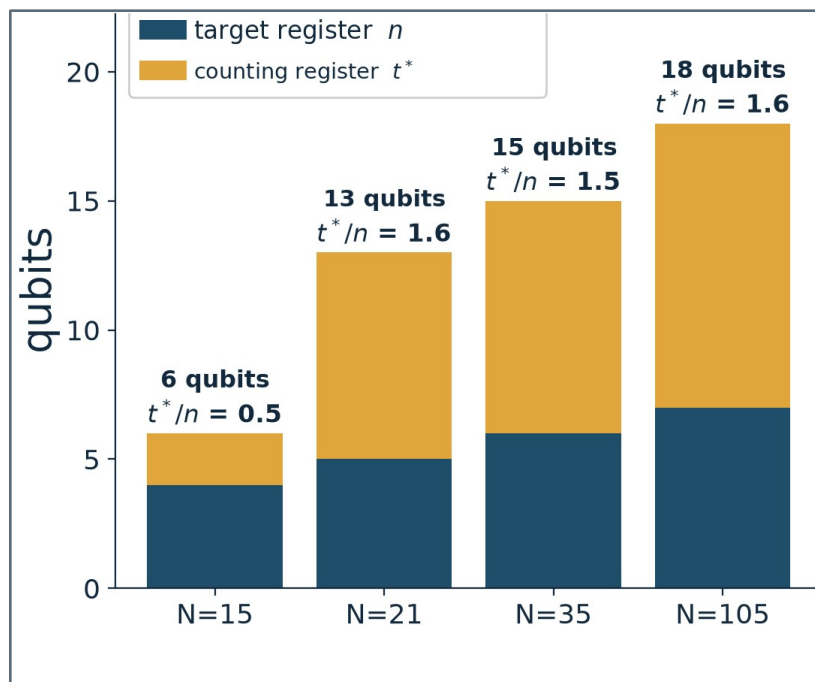
= $k = 0, r$ odd, $\gcd(k, r) > 1$. Constant in t , reduced by repeating (or changing a).

Separating the two is the point of the experiment.

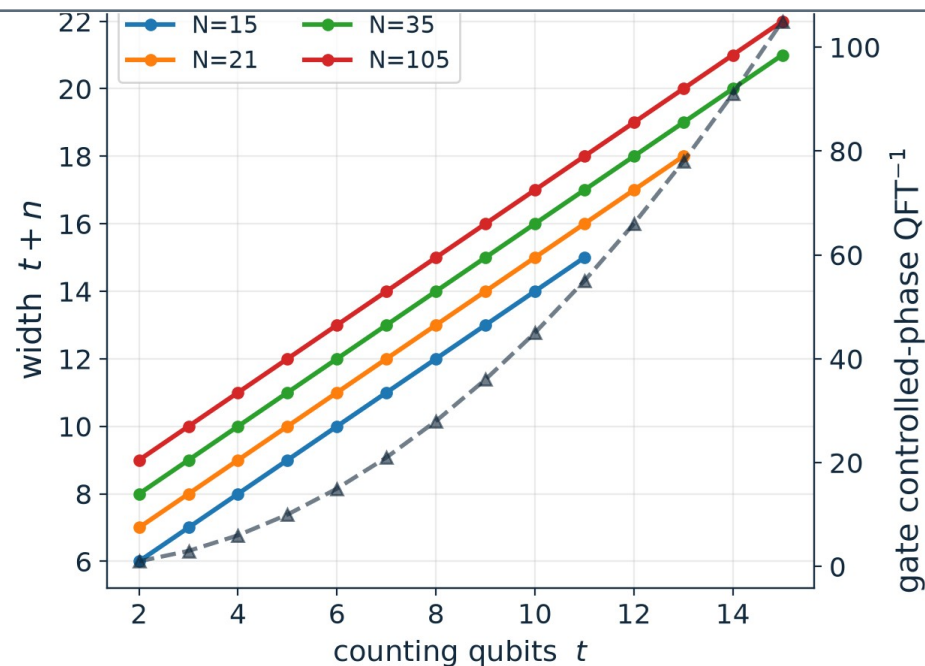
Let's define t^* as the smallest amount of counting qubits for which $P_{\text{factor}}(t)$ reaches 90% of its plateau value.

Width and circuit cost

Circuit width at t^*



Linear in width, quadratic in QFT^{-1}



$$t^* \approx 1.5-1.6 n$$

(N = 15: 0.5, the lucky case)

- width $t + n$: linear
- $\text{QFT}^{-1} t(t-1)/2$: quadratic
- only 2-4 distinct M_b operators

Conclusions

N	n	t^*	width	t^*/n
21	5	8	13	1.6
35	6	9	15	1.5
105	7	11	18	1.57

- The phase-only register is $\sim 1.5\times$ the one holding the number.
- Shor's theorem: asymptotically $t^* \sim n + m$, with $P_{\text{factor}} \geq 1 - \frac{1}{2}^m$ (m = extra qubits beyond n).
- Within the same circuit: linear width, quadratic QFT^{-1} , modular arithmetic with few distinct M_b .

References

- [1] **P. W. Shor**, *Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer*. SIAM J. Comput. 26(5), 1997 (arXiv: quant-ph/9508027).
- [2] **IBM Quantum**, *Shor's algorithm* — tutorial. quantum.cloud.ibm.com/docs/en/tutorials/shors-algorithm